

A 2-categorical approach to logical relations

FioreFest, Cambridge, June 2026

These slides available at philipsaville.co.uk

Pedro H. Azevedo de Amorim (University of Bath, UK)

Satoshi Kura (Waseda University, Japan)

Philip Saville (University of Sussex, UK)

Linus Schönfelder (University of Sussex, UK)



Things I learned from Marcelo

Things I learned from Marcelo

1. Look for the right
abstractions.

Things I learned from Marcelo

1. Look for the right
abstractions.

2. Let the algebra tell you
how things should be.

“correct-by-construction”

“principled”

“synthesise”

Logical relations,
at varying levels
of abstraction

Level 1: logical relations as families of relations Milner, Plotkin,

A **logical relation** R consists of:

a predicate $R_A \subseteq \llbracket A \rrbracket$ for each type A ...

...determined inductively at higher types by
a **logical relations condition**

Level 1: logical relations as families of relations Milner, Plotkin,

A **logical relation** R consists of:

a predicate $R_A \subseteq \llbracket A \rrbracket$ for each type A ...

...determined inductively at higher types by
a **logical relations condition**

E.g. logical relations condition for $\rightarrow$ in STLC:

$$f \in R_{A \rightarrow B} \subseteq \llbracket A \rrbracket \Rightarrow \llbracket B \rrbracket \quad \text{iff} \quad f(x) \in R_B \text{ whenever } x \in R_A$$

Level 1: logical relations as families of relations Milner, Plotkin,

A logical relation R consists of:

a predicate $R_A \subseteq \llbracket A \rrbracket$ for each type A ...

...determined inductively at higher types by
a logical relations condition

Basic Lemma:

every program is in the relation: $\llbracket P \rrbracket \in R_A$ for any closed $P : A$

so long as this holds for the basic constants

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

semantic model

Set

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

objects: $(X \in \mathbf{Set}, R \subseteq X)$

maps: functions preserving the predicate

Pred



semantic model

Set

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

objects: $(X \in \mathbf{Set}, R \subseteq X)$

maps: functions preserving the predicate

semantic model

Pred



Set

Pred is cartesian closed:

$$(X, R) \Rightarrow (Y, S) := (X \Rightarrow Y, R \supset S)$$

$$f \in R \supset S \text{ iff } f(x) \in S \text{ whenever } x \in R$$

$$f \in \llbracket A \rightarrow B \rrbracket^{\mathbf{Pred}} \text{ iff } f \text{ preserves the predicate on } \llbracket A \rrbracket^{\mathbf{Set}}$$

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

objects: $(X \in \mathbf{Set}, R \subseteq X)$
maps: functions preserving the predicate

semantic model

Pred



Set

Pred is cartesian closed:

$$(X, R) \Rightarrow (Y, S) := (X \Rightarrow Y, R \supset S)$$
$$f \in R \supset S \text{ iff } f(x) \in S \text{ whenever } x \in R$$

$$f \in \llbracket A \rightarrow B \rrbracket^{\mathbf{Pred}} \text{ iff } f \text{ preserves the predicate on } \llbracket A \rrbracket^{\mathbf{Set}}$$

CC-structure encodes the
logical relations condition!

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

objects: $(X \in \mathbf{Set}, R \subseteq X)$

maps: functions preserving the predicate

forgetful functor p

strictly preserves CC-structure

semantic model

Pred



Set

Pred is cartesian closed:

$$(X, R) \Rightarrow (Y, S) := (X \Rightarrow Y, R \supset S)$$

$$f \in R \supset S \text{ iff } f(x) \in S \text{ whenever } x \in R$$

$$f \in \llbracket A \rightarrow B \rrbracket^{\mathbf{Pred}} \text{ iff } f \text{ preserves the predicate on } \llbracket A \rrbracket^{\mathbf{Set}}$$

CC-structure encodes the
logical relations condition!

Level 2: logical relations as relations models

Statman,
Reynolds & Ma, ...

objects: $(X \in \mathbf{Set}, R \subseteq X)$
maps: functions preserving the predicate

forgetful functor p
strictly preserves CC-structure

semantic model

Pred



Set

Pred is cartesian closed:

$$(X, R) \Rightarrow (Y, S) := (X \Rightarrow Y, R \supset S)$$
$$f \in R \supset S \text{ iff } f(x) \in S \text{ whenever } x \in R$$

$$f \in \llbracket A \rightarrow B \rrbracket^{\mathbf{Pred}} \text{ iff } f \text{ preserves the predicate on } \llbracket A \rrbracket^{\mathbf{Set}}$$

Basic Lemma holds automatically:

$$p(\llbracket P \rrbracket^{\mathbf{Pred}}) = \llbracket P \rrbracket^{\mathbf{Set}}$$

CC-structure encodes the
logical relations condition!

Level 3: logical relations as certain fibrations

Hermida, Jacobs, Katsumata, ...

Level 3: logical relations as certain fibrations

Hermida, Jacobs, Katsumata,

semantic model

$\mathcal{M}$

Level 3: logical relations as certain fibrations

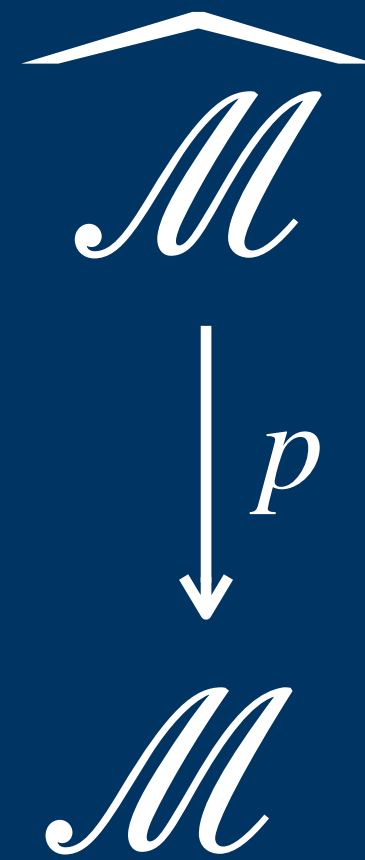
Hermida, Jacobs, Katsumata,

‘relations model’

forgetful functor p

strictly preserves model-structure

semantic model



Level 3: logical relations as certain fibrations

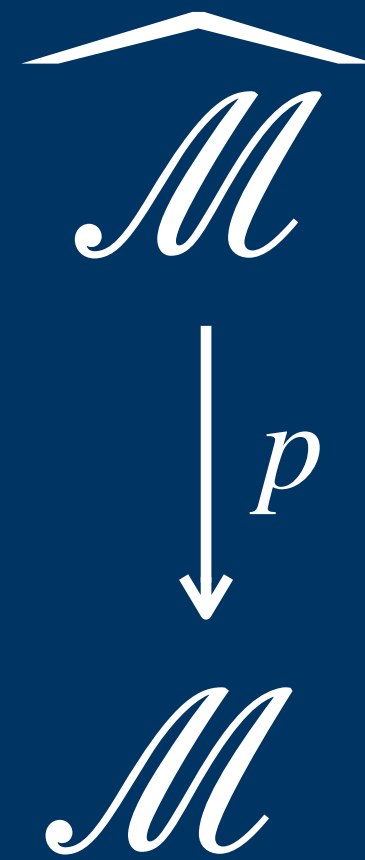
Hermida, Jacobs, Katsumata,

‘relations model’ = **fibration**

forgetful functor p

strictly preserves model-structure

semantic model



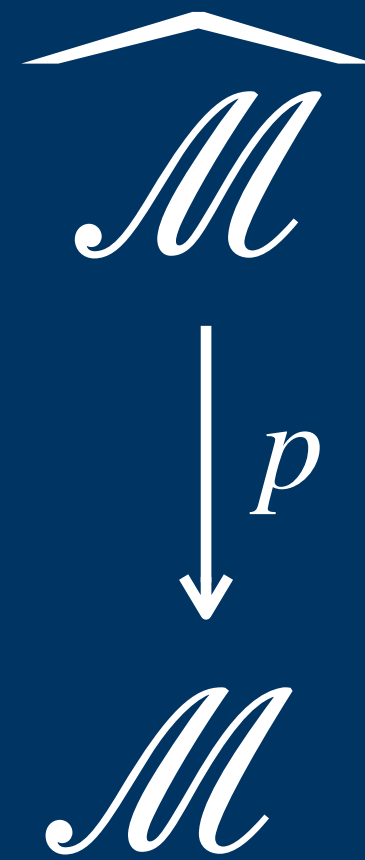
Level 3: logical relations as certain fibrations

Hermida, Jacobs, Katsumata,

‘relations model’ = **fibration**

forgetful functor p
strictly preserves model-structure

semantic model



model-structure encodes the
logical relations condition

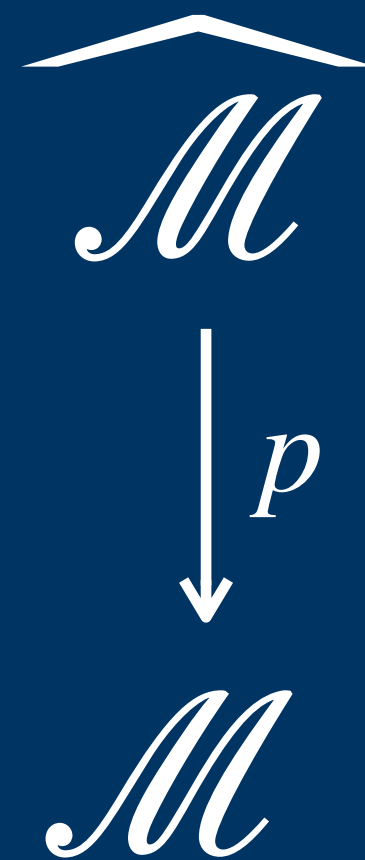
Level 3: logical relations as certain fibrations

Hermida, Jacobs, Katsumata,

‘relations model’ = **fibration**

forgetful functor p
strictly preserves model-structure

semantic model



model-structure encodes the
logical relations condition

Basic Lemma holds automatically:

$$p\left(\llbracket P \rrbracket^{\widehat{\mathcal{M}}}\right) = \llbracket P \rrbracket^{\mathcal{M}}$$

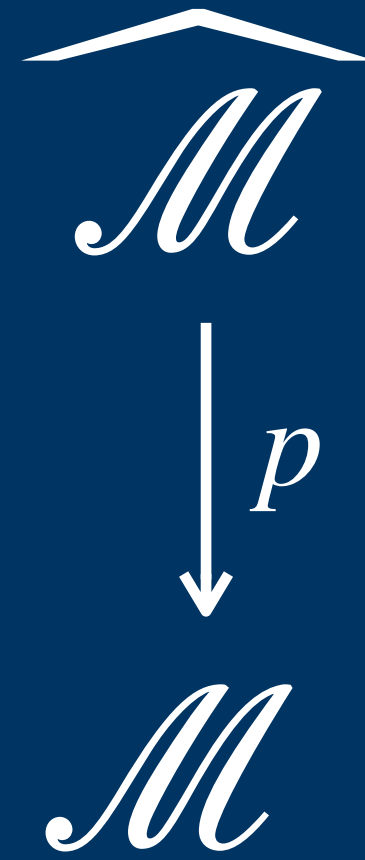
Level 3: logical relations as certain fibrations

Hermida, Jacobs, Katsumata,

‘relations model’ = **fibration**

forgetful functor p
strictly preserves model-structure

semantic model



model-structure encodes the
logical relations condition

Basic Lemma holds automatically:

$$p(\llbracket P \rrbracket^{\widehat{\mathcal{M}}}) = \llbracket P \rrbracket^{\mathcal{M}}$$

fibration for logical relations:

fibration
strictly preserving
model structure

Logical relations can be seen as...

Level 1: type-indexed families of relations

Level 2: relations models

Level 3: fibrations for logical relations

fibration for logical relations:

fibration

strictly preserving

model structure

Logical relations can be seen as...

Level 1: type-indexed families of relations

Level 2: relations models

Level 3: fibrations for logical relations



also get:
a good theory for building
new logical relations

fibration for logical relations:
fibration
strictly preserving
model structure

Building fibrations for logical relations

e.g. effect simulation, Π -lifting, Kripke relations of varying arity, ...

fibration for logical relations:

fibration
strictly preserving
model structure

Building fibrations for logical relations

e.g. effect simulation, TT -lifting, Kripke relations of varying arity, ...

fibration for logical relations:

fibration
strictly preserving
model structure

semantic
model

$\mathcal{M}$

Building fibrations for logical relations

e.g. effect simulation, TT -lifting, Kripke relations of varying arity, ...

fibration for logical relations:

fibration
strictly preserving
model structure

semantic
model

$\mathcal{M}$

$\mathcal{E}$

$\downarrow p$

$\mathcal{B}$

abstract notion
of ‘relation’

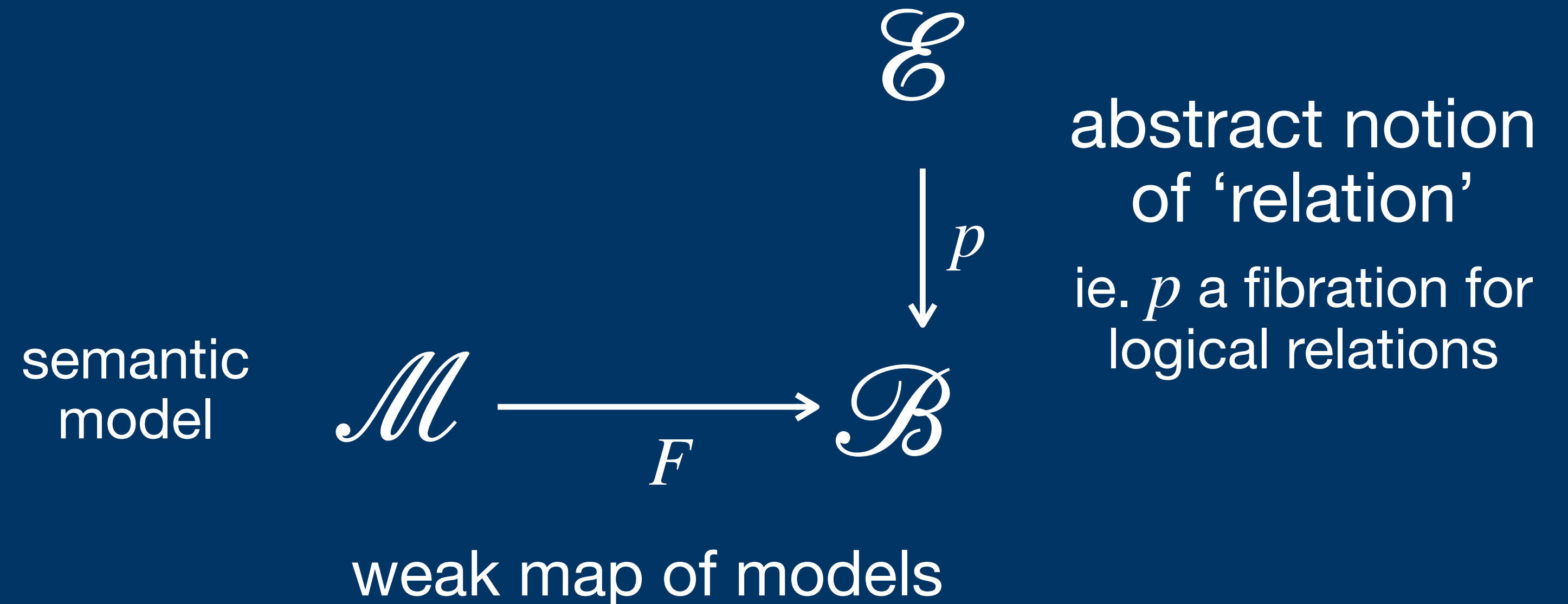
ie. p a fibration for
logical relations

Building fibrations for logical relations

e.g. effect simulation, Π -lifting, Kripke relations of varying arity, ...

fibration for logical relations:

fibration
strictly preserving
model structure



Building fibrations for logical relations

e.g. effect simulation, Π -lifting, Kripke relations of varying arity, ...

fibration for logical relations:

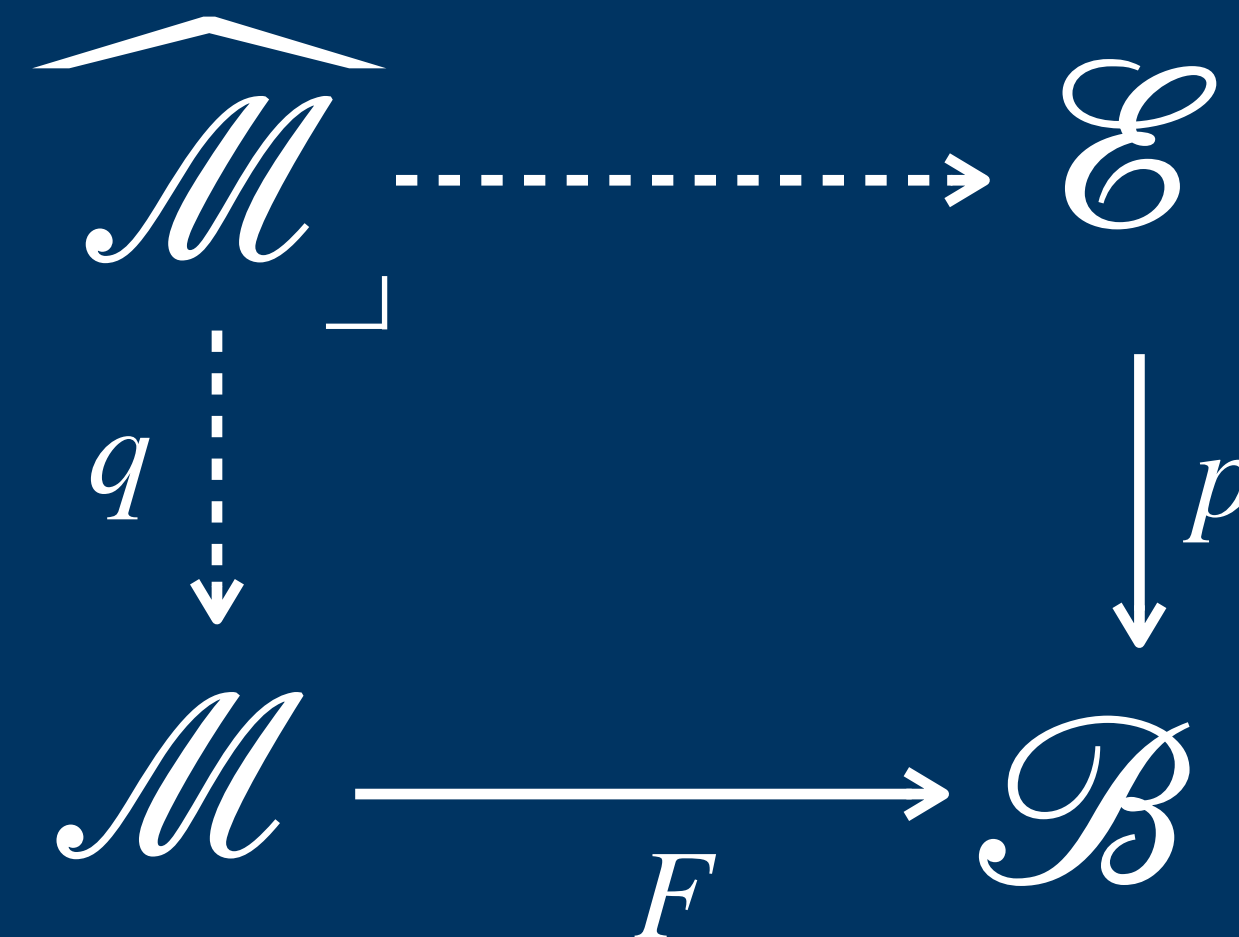
fibration
strictly preserving
model structure

lifting theorem:

universal choice of model

‘glueing’ objects of $\mathcal{M}$ to ‘relations’ in $\mathcal{E}$

semantic
model



weak map of models

abstract notion
of ‘relation’

ie. p a fibration for
logical relations

Building fibrations for logical relations

e.g. effect simulation, Π -lifting, Kripke relations of varying arity, ...

lifting theorem:

universal choice of model
'glueing' objects of $\mathcal{M}$ to 'relations' in $\mathcal{E}$

fibration for logical relations:

fibration
strictly preserving
model structure

objects: $(X \in \mathbf{Set}, R \subseteq X \times X)$

maps: pairs of maps preserving the relation

$$\begin{array}{ccc} \mathbf{BinPred} & \dashrightarrow & \mathbf{Pred} \\ \downarrow q & \lrcorner & \downarrow p \\ \mathbf{Set} \times \mathbf{Set} & \xrightarrow{\times} & \mathbf{Set} \end{array}$$

$$\begin{array}{ccc} \widehat{\mathcal{M}} & \dashrightarrow & \mathcal{E} \\ \downarrow q & \lrcorner & \downarrow p \\ \mathcal{M} & \xrightarrow{F} & \mathcal{B} \end{array}$$

weak map of models

abstract notion
of 'relation'

ie. p a fibration for
logical relations

Logical relations can be seen as...

Level 1: type-indexed families of relations

Level 2: relations models

Level 3: fibrations for logical relations



+ can build new examples
from old ones by glueing

fibration for logical relations:
fibration
strictly preserving
model structure

Logical relations: a 2-categorical perspective

What's a fibration for logical relations for call-by-push-value?

A **CBPV-model** is a $\mathbf{Psh}(\mathcal{V})$ -enriched adjunction

$$\mathbf{self} \mathcal{V} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow[U]{\perp} \end{array} \mathcal{C}$$

1. What is a **CBPV** fibration for logical relations?
2. Do they satisfy a **lifting theorem**?

What's a fibration for logical relations for call-by-push-value?

A **CBPV-model** is a $\mathbf{Psh}(\mathcal{V})$ -enriched adjunction

$$\mathbf{self} \mathcal{V} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow[U]{\perp} \end{array} \mathcal{C}$$

1. What is a **CBPV** fibration for logical relations?
2. Do they satisfy a **lifting theorem**?

can no longer just say a fibration which preserves all the model structure

need a sensible notion of glueing

Observation:

“fibration which preserves model structure”	$\approx$	fibration internal to 2-category of models
---	-----------	--

A fibration internal to...

is...

CartCart_{strict}

a fibration strictly preserving products

CCCart_{strict}

a fibration strictly preserving CC-structure

λ_{ml} -models

a fibration strictly preserving λ_{ml} -structure

Suggestion for level 4:

fibration for logical relations	$\coloneqq$	fibration internal to 2-category of models
------------------------------------	-------------	--

*sometimes sufficient but not necessary condition, but a lot of the theory still applies

Suggestion for level 4:

fibration for logical relations	$\coloneqq$	fibration internal to 2-category of models
------------------------------------	-------------	--

*sometimes sufficient but not necessary condition, but a lot of the theory still applies

Does this work correctly?

Internal fibrations are always fibrations which preserve model structure

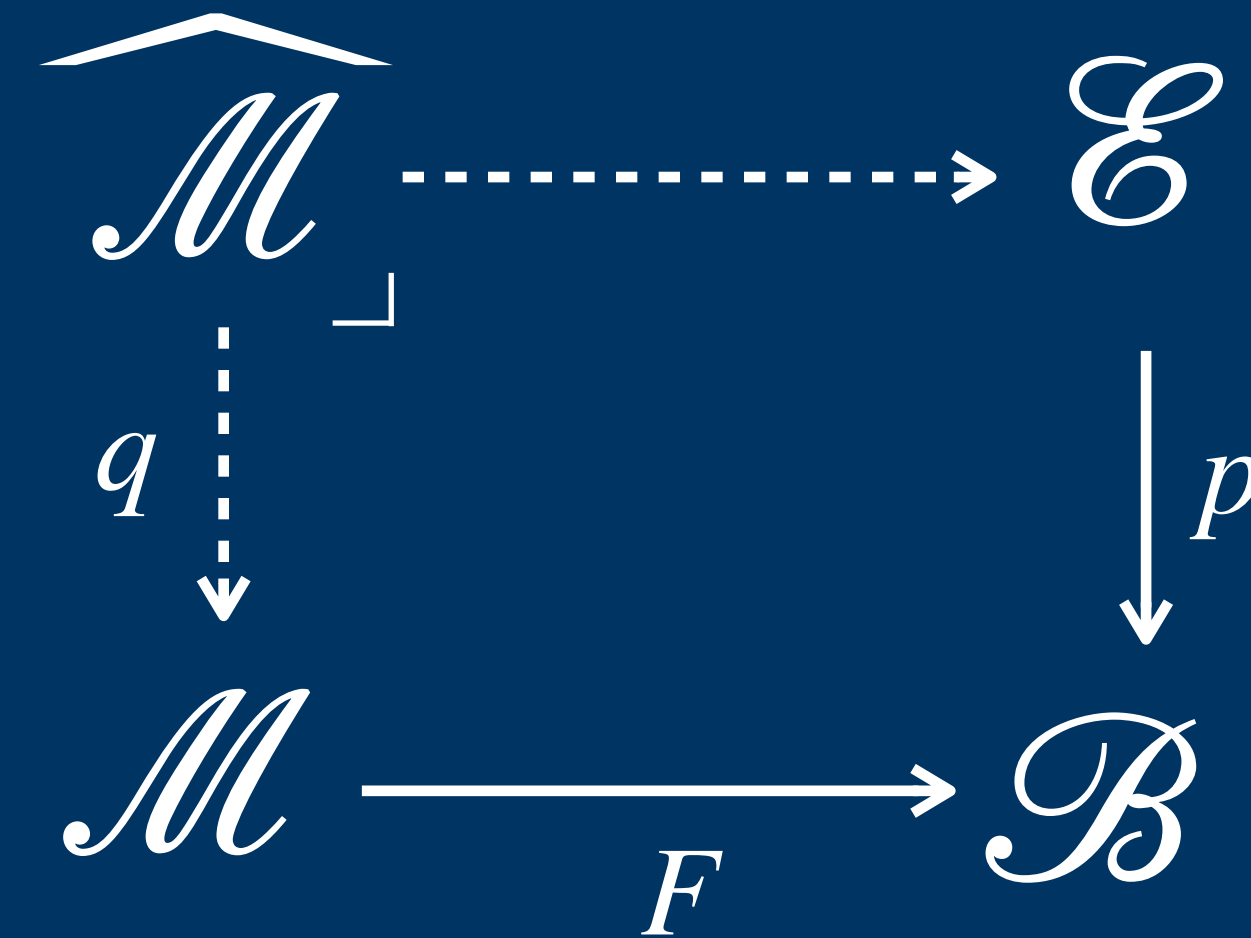
Theorem: let T be a relative pseudomonad along a pseudofunctor $J : \mathcal{A} \rightarrow \mathcal{C}$. Then a fibration in $\text{Ps-}T\text{-Alg}_{\text{ps}}$ is exactly an algebra pseudomorphism $(p, \bar{p})$ such that p is a fibration in $\mathcal{C}$.

Glueing plays well with fibrations

lifting theorem:

universal choice of model
'glueing' objects of $\mathcal{M}$ to 'relations' in $\mathcal{E}$

semantic
model



weak map of models

abstract notion
of 'relation'
ie. p a fibration for
logical relations

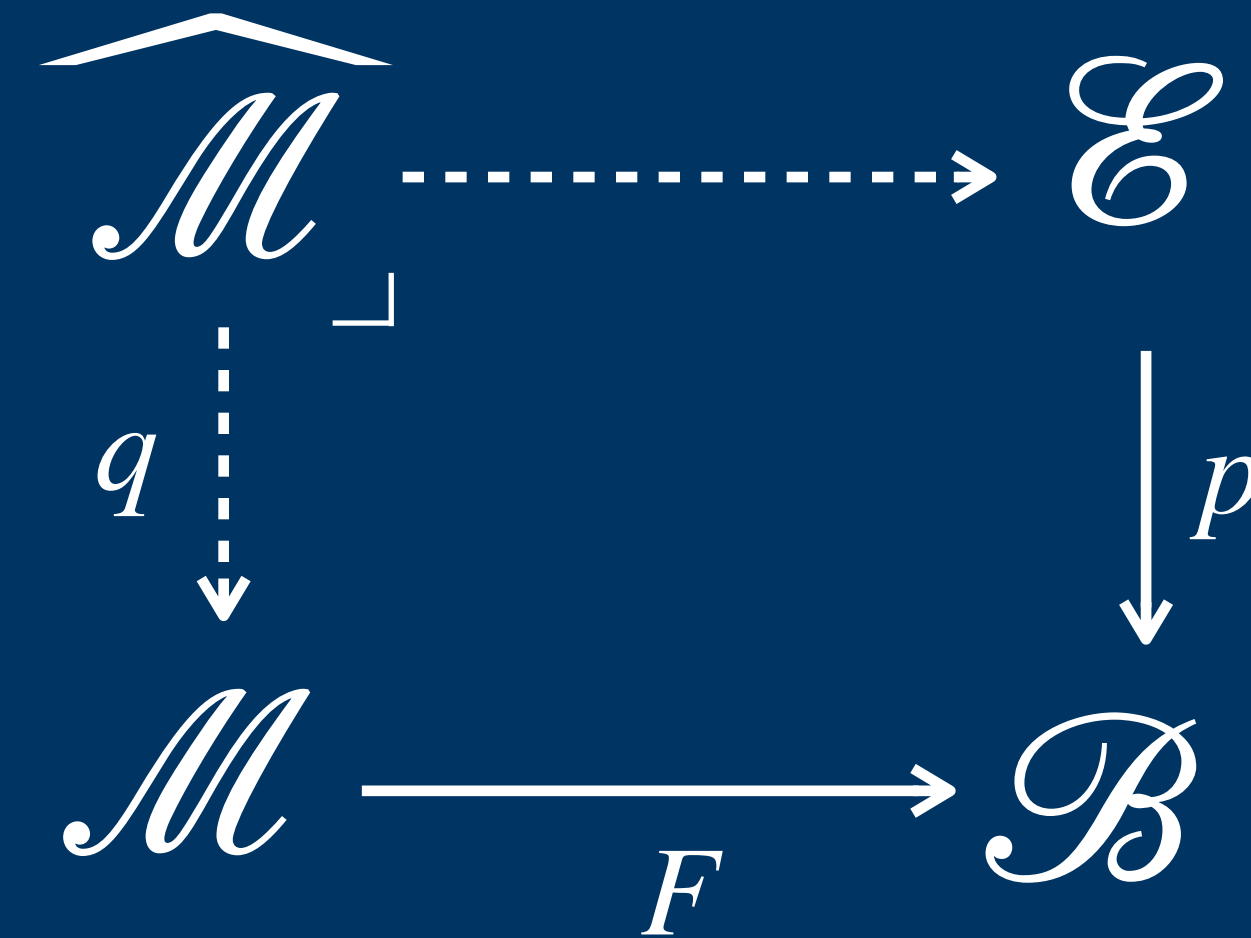
Proposition [classic]: fibrations are closed under pullback.

Glueing plays well with fibrations

lifting theorem:

universal choice of model
'glueing' objects of $\mathcal{M}$ to 'relations' in $\mathcal{E}$

semantic
model



abstract notion
of 'relation'
ie. p a fibration for
logical relations

weak map of models

Proposition [classic]: fibrations are closed under pullback.

Glueing reduces to the existence of certain pullbacks in 2-categories of models.

Glueing plays well with fibrations

Proposition: let T be a relative pseudomonad. Consider the following cospan of pseudoalgebras:

$$\begin{array}{ccc} & (E, \dots) & \\ & \downarrow (p, \bar{p}) & \text{lax algebra map +} \\ (A, \dots) & \xrightarrow[(f, \bar{f})]{\text{oplax algebra map}} & (B, \dots) \end{array}$$

fibration

Glueing plays well with fibrations

Proposition: let T be a relative pseudomonad. Consider the following cospan of pseudoalgebras:

$$\begin{array}{ccc}
 (P, \dots) & \dashrightarrow & (E, \dots) \\
 \downarrow (f^*(p), \text{id}) & & \downarrow (p, \bar{p}) \\
 (A, \dots) & \xrightarrow[(f, \bar{f})]{\text{oplax algebra map}} & (B, \dots)
 \end{array}
 \quad \begin{array}{l} \\ \text{lax algebra map +} \\ \text{fibration} \end{array}$$

Then for the pullback $f^*(p) : P \rightarrow A$ in $\mathcal{C}$:

1. P acquires oplax algebra structure,
2. $f^*(p)$ is a strict algebra map, and
3. If $(p, \bar{p})$ is pseudo, this is a 2-pullback in $\text{Oplax-}T\text{-Alg}_{\text{oplax}}$.

Glueing plays well with fibrations
...even when your models are diagrams

Glueing plays well with fibrations

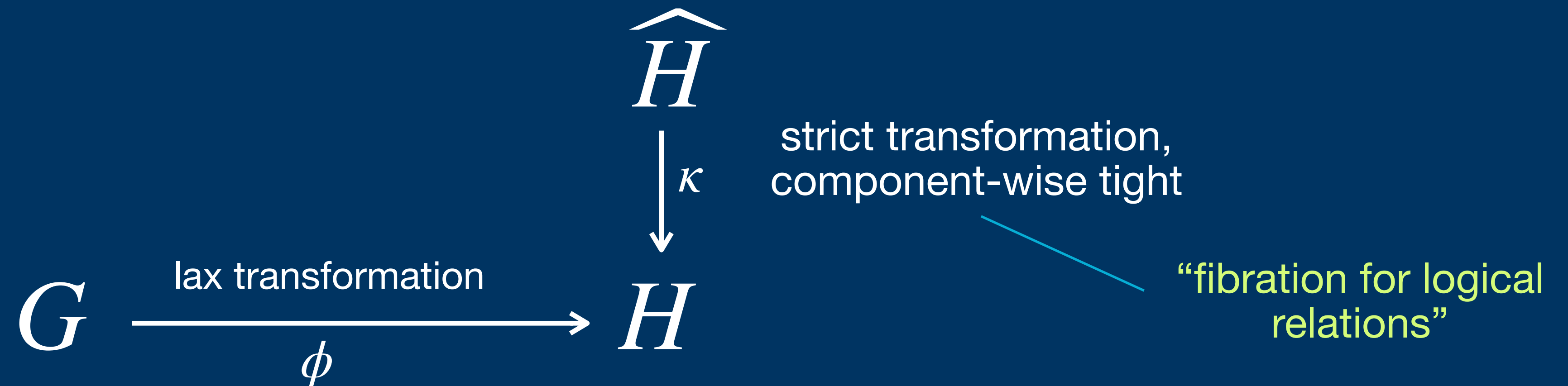
...even when your models are diagrams

If **tight** maps in $\mathcal{C}$ are closed under pullback (+ satisfy mild conditions), then in $[\mathcal{I}, \mathcal{C}]_{\text{fax}}$ you can pull a tight map back along any map:

Glueing plays well with fibrations

...even when your models are diagrams

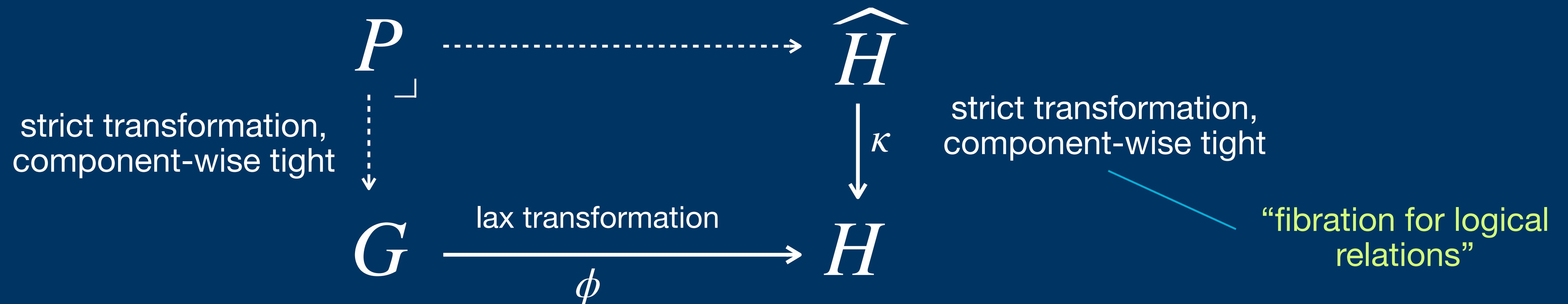
If **tight** maps in $\mathcal{C}$ are closed under pullback (+ satisfy mild conditions), then in $[\mathcal{I}, \mathcal{C}]_{\text{lax}}$ you can pull a tight map back along any map:



Glueing plays well with fibrations

...even when your models are diagrams

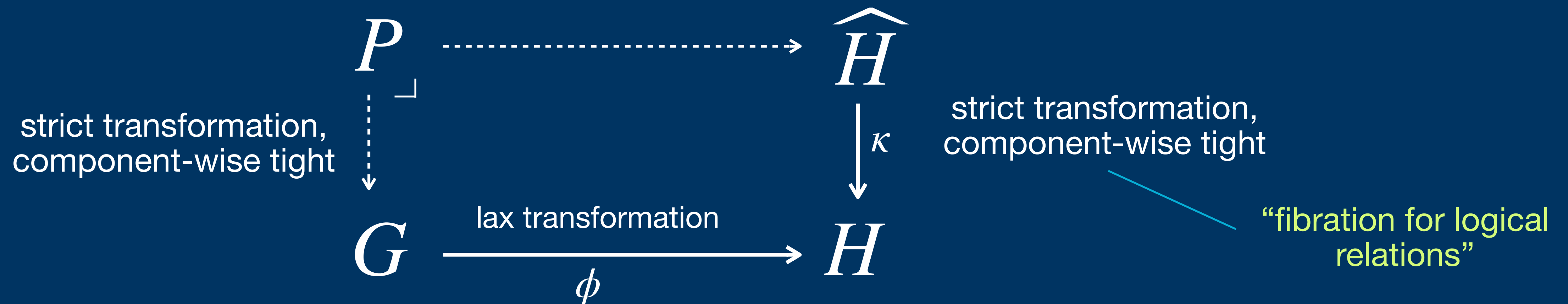
If **tight** maps in $\mathcal{C}$ are closed under pullback (+ satisfy mild conditions), then in $[\mathcal{I}, \mathcal{C}]_{\text{lax}}$ you can pull a tight map back along any map:



Glueing plays well with fibrations

...even when your models are diagrams

If **tight** maps in $\mathcal{C}$ are closed under pullback (+ satisfy mild conditions), then in $[\mathcal{I}, \mathcal{C}]_{\text{lax}}$ you can pull a tight map back along any map:



Moral: glueing works for monad models and adjunction models.

under slightly stronger conditions:
can also take fibred products à la
TT-lifting with multiple parameters

1. Fibrations internal to 2-categories tend to be the right thing*.
2. The theory of glueing works as you would hope, even in 2-categories of diagrams.

*sometimes sufficient but not necessary condition, but a lot of the theory still applies

1. Fibrations internal to 2-categories tend to be the right thing*.
2. The theory of glueing works as you would hope, even in 2-categories of diagrams.

*sometimes sufficient but not necessary condition, but a lot of the theory still applies

~ suggests a recipe for approaching logical relations denotationally

Logical relations for CBPV models

(a recipe)

1. Define a 2-category $\mathbf{LInd}$ of locally-indexed categories,
2. Define CBPV as a sub-2-category of $[\mathbf{Adj}, \mathbf{LInd}]_{\mathbf{Lax}}$,
3. Identify the fibrations internal to $\mathbf{LInd}$,
4. Define a CBPV fibration to be a CBPV-morphism that's component-wise a $\mathbf{LInd}$ -fibration.

logical relations models + a glueing theorem follow from the algebra

including codomain fibration, subobject fibration, versions of TT-lifting, ...

a.k.a. Kripke relations of varying arity



Next steps: open logical relations

characterise the definable
maps in a model
[cf. Jung & Tiuryn]

semantic NbE à la
*Semantic Analysis of
Normalisation by Evaluation*

a.k.a. Kripke relations of varying arity



Next steps: open logical relations

characterise the definable
maps in a model
[cf. Jung & Tiuryn]

semantic NbE à la
*Semantic Analysis of
Normalisation by Evaluation*

Missing ingredient: a **presheaf** construction for CBPV models

2-categorical statement
of Lafont argument

Open logical relations

(a.k.a. Kripke relations of varying arity)

Many properties of interest require us to pay attention to **contexts**.

For example: definability (Jung & Tiuryn)

$$\text{Def}_\sigma(\Gamma) := \{ \llbracket M \rrbracket \mid \Gamma \vdash M : \sigma \} \subseteq \mathcal{M}(\llbracket \Gamma \rrbracket, \llbracket \sigma \rrbracket)$$

Open logical relations

(a.k.a. Kripke relations of varying arity)

Many properties of interest require us to pay attention to **contexts**.

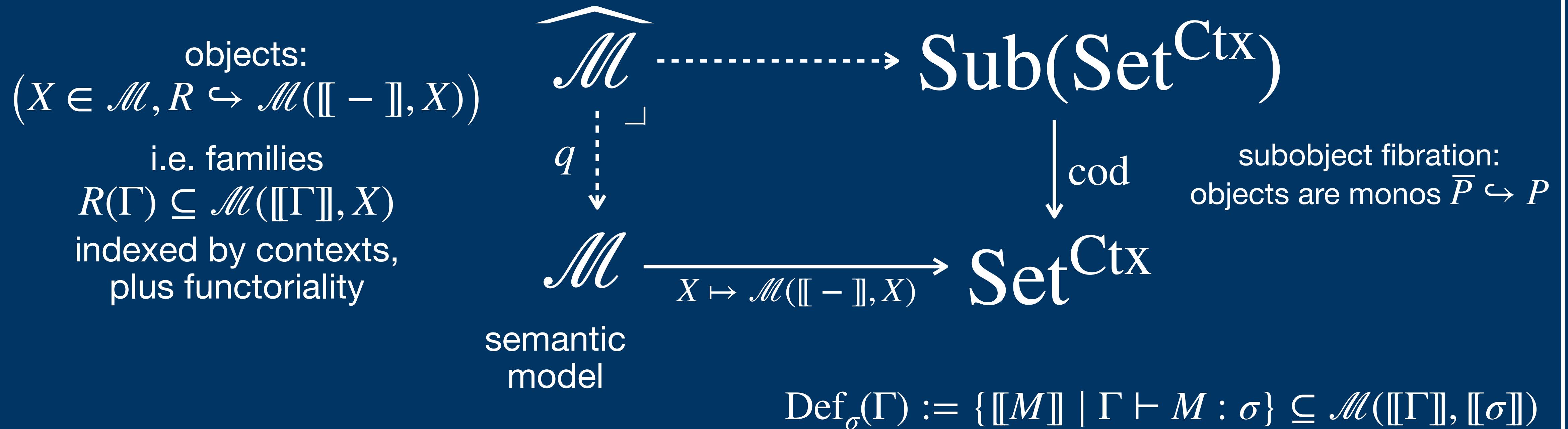
For example: definability (Jung & Tiuryn)

$$\text{Def}_\sigma(\Gamma) := \{ \llbracket M \rrbracket \mid \Gamma \vdash M : \sigma \} \subseteq \mathcal{M}(\llbracket \Gamma \rrbracket, \llbracket \sigma \rrbracket)$$

$$\llbracket \lambda x . M \rrbracket \in \text{Def}_{\sigma \rightarrow \tau}(\Gamma) \iff \llbracket M \rrbracket \in \text{Def}_\tau(\Gamma, x : \sigma)$$

Open logical relations

(a.k.a. Kripke relations of varying arity)



open logical relations = glueing + presheaves



???

What about call-by-push-value?

= “locally indexed”

A **CBPV-model** is a $\mathbf{Psh}(\mathcal{V})$ -enriched adjunction

$$\mathbf{self} \mathcal{V} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow[\quad]{\perp} \\ \xleftarrow{U} \end{array} \mathcal{C}$$

What is a ‘presheaf model’ for CBPV?

What about call-by-push-value?

= “locally indexed”

A **CBPV-model** is a $\mathbf{Psh}(\mathcal{V})$ -enriched adjunction

$$\mathbf{self} \mathcal{V} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow[\quad]{\perp} \\ \xleftarrow{U} \end{array} \mathcal{C}$$

What is a ‘presheaf model’ for CBPV?

First approach: use the enriched presheaf construction.
But: size issues become troubling.

What about call-by-push-value?

= “locally indexed”

A **CBPV-model** is a $\mathbf{Psh}(\mathcal{V})$ -enriched adjunction

$$\mathbf{self} \mathcal{V} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow[\quad]{\perp} \\ \xleftarrow{U} \end{array} \mathcal{C}$$

What is a ‘presheaf model’ for CBPV?

First approach: use the enriched presheaf construction.
But: size issues become troubling.

Alternative: recast CBPV models as certain **fibrations**.

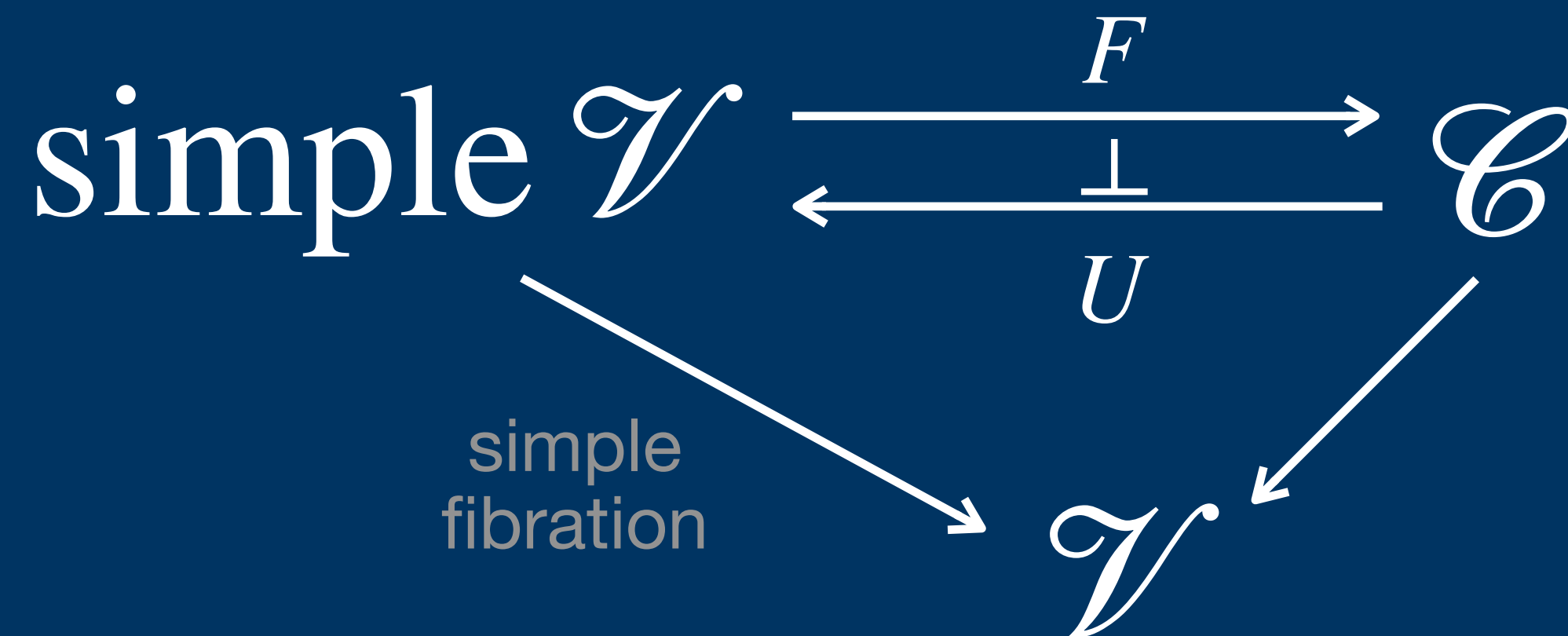
(cf. Ahman, Vákár)

What about call-by-push-value?

each fibre has the same
set of objects, and
reindexing is *ioo*

$\text{Psh}(\mathcal{V})$ -enriched category $\mathcal{C}$ $\overset{\text{folklore(?)}}{\longleftrightarrow}$ ‘local’ fibration
 $p : \mathcal{E} \rightarrow \mathcal{V}$

A **CBPV-model** is a (locally) fibred adjunction:



(cf. Ahman, Vákár)

The 2-category of local fibrations has a Yoneda structure

$$\begin{array}{ccccc}
 \mathcal{E} & & \mathcal{P}(p) & \dashrightarrow & \text{simple}(\text{Psh}(\mathcal{E})) \\
 \downarrow p & \mapsto & \downarrow \lrcorner & & \downarrow \text{simple fibration} \\
 \mathcal{B} & & \text{Psh}(\mathcal{B}) & \xrightarrow{p^*} & \text{Psh}(\mathcal{E})
 \end{array}$$

Induces a presheaf pseudofunctor $\leadsto$ sends adjunctions to adjunctions.

Determines a canonical **presheaf operation on CBPV models**.

Extracting the juice (WIP)

1. Open logical relations for CBPV: characterising definability, semantic NbE.
2. 2-categorical Lafont argument using this construction.
3. Seems to be quite a general way to build presheaf-like pseudomonads.

Summing up

1. Theory of logical relations includes **fibrations for logical relations** and **glueing**.
2. Fibrations **internal to 2-categories of models** seem to capture both aspects quite well.
3. Applies to examples where the models are more complex, e.g. CBPV.
4. Defining an appropriate **presheaf construction** should also lead to open logical relations, definability, NbE, conservativity,

the right
abstraction (?)



follow what
the algebra
tells you



Summing up

1. Theory of logical relations includes **fibrations for logical relations** and **glueing**.
2. Fibrations **internal to 2-categories of models** seem to capture both aspects quite well.
3. Applies to examples where the models are more complex, e.g. CBPV.
4. Defining an appropriate **presheaf construction** should also lead to open logical relations, definability, NbE, conservativity,

the right
abstraction (?)



follow what
the algebra
tells you



Conjecture: you can play the same game for other interesting examples.

Summing up

1. Theory of logical relations includes **fibrations for logical relations** and **glueing**.
2. Fibrations **internal to 2-categories of models** seem to capture both aspects quite well.
3. Applies to examples where the models are more complex, e.g. CBPV.
4. Defining an appropriate **presheaf construction** should also lead to open logical relations, definability, NbE, conservativity,

the right
abstraction (?)



follow what
the algebra
tells you



Conjecture: you can play the same game for other interesting examples.

Suggests plenty of other applications + a ‘formal theory of logical relations’.